

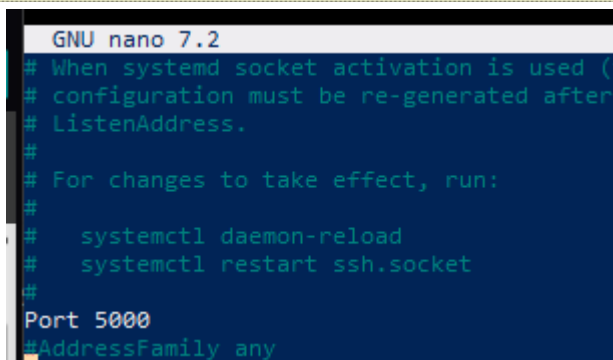
U1. PRÁCTICAS Acceso y control remoto

1. Configura un servidor ssh que tenga la siguiente configuración:

- Que el puerto de escucha del servidor ssh se establezca en el puerto 5000.

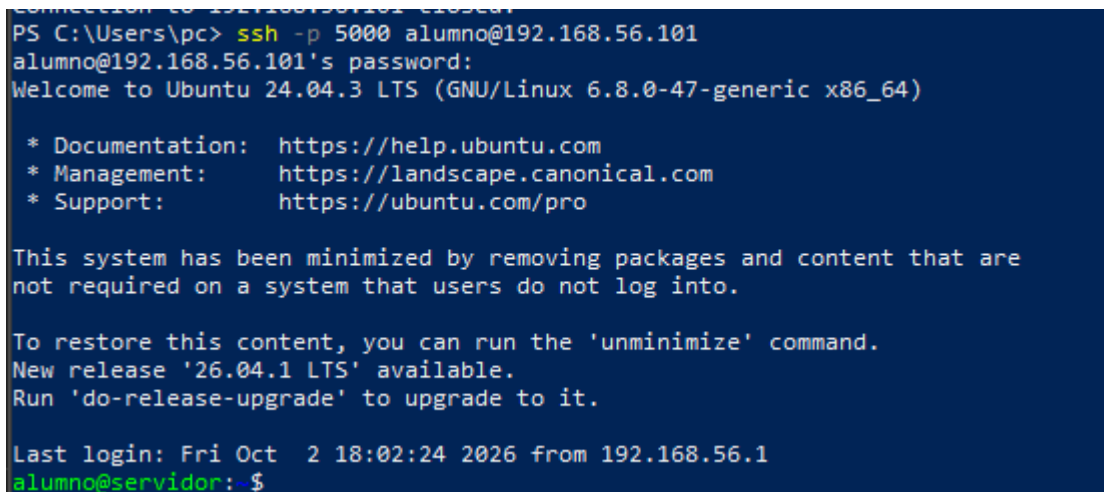
Fichero de configuración. Escribe la línea que has modificado en el fichero de configuración:

Port 5000



```
GNU nano 7.2
# When systemd socket activation is used (t
# configuration must be re-generated after
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 5000
#AddressFamily any
```

Captura de pantalla que demuestre que funciona vuestra configuración.



```
connection to 192.168.56.101 closed.
PS C:\Users\pc> ssh -p 5000 alumno@192.168.56.101
alumno@192.168.56.101's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-47-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Oct  2 18:02:24 2026 from 192.168.56.1
alumno@servidor: $
```

- Crea un usuario llamado prueba en tu servidor pero que no pueda acceder mediante al servidor mediante el servicio ssh.

Fichero de configuración. Escribe la línea que has modificado en el fichero de configuración:

DenyUsers prueba

```
GNU nano 7.2
# When systemd socket activation is used (the d
# configuration must be re-generated after chan
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 5000
DenyUsers prueba
#AddressFamily any
```

Captura de pantalla que demuestre que funciona vuestra configuración.

```
connection to 192.168.56.101 closed.
PS C:\Users\pc> ssh -p 5000 prueba@192.168.56.101
prueba@192.168.56.101's password: 
Permission denied, please try again.
prueba@192.168.56.101's password: 
```

- Haz que sólo el usuario llamado prueba pueda acceder mediante al servidor mediante el servicio ssh. El resto de usuarios no podrán acceder

Fichero de configuración. Escribe la línea que has modificado en el fichero de configuración:

AllowUsers prueba

```
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 5000
AllowUsers prueba
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
```

```
alumno@servidor:~$ sudo systemctl restart ssh
alumno@servidor:~$ exit
logout
Connection to 192.168.56.101 closed.
PS C:\Users\pc> ssh -p 5000 alumno@192.168.56.101
alumno@192.168.56.101's password: 
Permission denied, please try again.
alumno@192.168.56.101's password: 
```

- Impide que se pueda conectar remotamente el usuario root.

Fichero de configuración. Escribe la línea que has modificado en el fichero de configuración:

PermitRootLogin no

```
# Authentication:
#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
```

Captura de pantalla que demuestre que funciona vuestra configuración.

```
logout
Connection to 192.168.56.101 closed.
PS C:\Users\pc> ssh -p 5000 root@192.168.56.101
root@192.168.56.101's password:
Permission denied, please try again.
root@192.168.56.101's password:
```

- No permitir conexiones a usuarios con contraseñas vacías.

Fichero de configuración. Escribe la línea que has modificado en el fichero de configuración:

PermitEmptyPasswd no

```
# To disable tunneled clear text passwords, change to
#PasswordAuthentication yes
PermitEmptyPasswords no
```

Captura de pantalla que demuestre que funciona vuestra configuración.

```
Connection to 192.168.56.101 closed.
PS C:\Users\pc> ssh -p 5000 alumno@192.168.56.101
alumno@192.168.56.101's password:
Permission denied, please try again.
alumno@192.168.56.101's password: █
```

2. Copia de fichero mediante ssh.

- Utiliza el comando scp para copiar un fichero llamado copiaprueba de tu máquina cliente a tu servidor ssh.

Comando utilizado:

```
scp C:\Users\pc\Downloads\copiaprueba1.odp
alumno@192.168.56.101:/home/alumno
```

```
PS C:\Users\pc> scp C:\Users\pc\Downloads\copiaprueba1.odp alumno@192.168.56.101:/home/alumno/
alumno@192.168.56.101's password:
copiaprueba1.odp 100% 668 217.5KB/s 00:00
PS C:\Users\pc>
```

Captura de pantalla que demuestre que funciona vuestra configuración.

```
alumno@servidor:~$ ls
copiaprueba1.odp
alumno@servidor:~$
```

- Utiliza el comando scp para copiar un fichero llamado copiaprueba2 de tu máquina cliente a tu servidor ssh.

Comando utilizado:

```
scp C:\Users\pc\Downloads\copiaprueba1.odp  
alumno@192.168.56.101:/home/alumno
```

```
PS C:\Users\pc> scp C:\Users\pc\Downloads\copiaprueba2.odp alumno@192.168.56.101:/home/alumno/  
alumno@192.168.56.101's password:  
copiaprueba2.odp 100% 668 163.1KB/s 00:00  
PS C:\Users\pc>
```

Captura de pantalla que demuestre que funciona vuestra configuración.

```
Last login: Fri Oct 2 18:31:03 2026 from 192.168.56.1  
alumno@servidor:~$ ls  
copiaprueba1.odp copiaprueba2.odp  
alumno@servidor:~$
```

3. Configura el cliente para que tenga un par de claves ssh que permitan el acceso mediante clave pública del cliente y no te pida la contraseña.

Fichero de configuración. Escribe la linea que has modificado en el fichero de configuración:

Captura de pantalla que demuestre que funciona vuestra configuración. Que el usuario accede al servidor ssh sin que te pida la contraseña.

```

pc-jsantos@DESKTOP-C1743AU:/mnt/c/Users/pc$ ssh-keygen
Command 'ssh-keygen' not found, did you mean:
  command 'ssh-keygen' from deb openssh-client (1:9.6p1-3ubuntu13.16)
Try: sudo apt install <deb name>
pc-jsantos@DESKTOP-C1743AU:/mnt/c/Users/pc$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/pc-jsantos/.ssh/id_ed25519):
Created directory '/home/pc-jsantos/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/pc-jsantos/.ssh/id_ed25519
Your public key has been saved in /home/pc-jsantos/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:aWlZAbDsS4WNzCG6K3L1uABZoxwre+WHGAN7lqoXfXY pc-jsantos@DESKTOP-C1743AU
The key's randomart image is:
+--[ED25519 256]--+
|  . o..          |
|  . = * .        |
| o.o B o .       |
| .Bo+ . o o      |
| Bo=. * + S      |
| *.X.*ooE        |
| = *.oo.         |
| O+.. o          |
| .. .           |
+-----[SHA256]-----+
pc-jsantos@DESKTOP-C1743AU:/mnt/c/Users/pc$ ssh-copy-id alumno@192.186.56.101
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/pc-jsantos/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: ERROR: ssh: connect to host 192.186.56.101 port 22: Connection refused

pc-jsantos@DESKTOP-C1743AU:/mnt/c/Users/pc$ ssh-copy-id alumno@192.168.56.101
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/pc-jsantos/.ssh/id_ed25519.pub"
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.
ED25519 key fingerprint is SHA256:FOYodorlei7Ax/eJ4uIoKB1FgwBnEYCuI50VM0u6UuM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
alumno@192.168.56.101's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'alumno@192.168.56.101'"
and check to make sure that only the key(s) you wanted were added.

pc-jsantos@DESKTOP-C1743AU:/mnt/c/Users/pc$ ssh alumno@192.168.56.101
Enter passphrase for key '/home/pc-jsantos/.ssh/id_ed25519':
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-47-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Oct  2 18:32:17 2026 from 192.168.56.1
alumno@servidor: $

```

PD: no sé porque no me ha aparecido la ventana como la que nos aparece en clase. En vez de eso me lo ha pedido mediante consola. Supongo que se debe a hacerlo en Windows

4. Configura tu servidor SSH para que admita doble autenticación mediante el servicio de google de Authenticator.

Captura de pantalla que muestre el acceso por ssh con doble autenticación.

GNU nano 7.2

PAM configuration for the Secure Shell service

Standard Un*x authentication.

@include common-auth

#One-time password authentication via Google Authenticator

auth required pam_google_authenticator.so

Change to yes to enable challenge-response passwords (beware issues with
some PAM modules and threads)

KbdInteractiveAuthentication yes

No VM guests are running outdated hypervisor (qemu) binaries on this host.

alumno@servidor: \$ google-authenticator

Do you want authentication tokens to be time-based (y/n) y

Warning: pasting the following URL into your browser exposes the OTP secret to Google:

<https://www.google.com/chart?chs=200x200&chld=M|0&cht=qr&chl=otpauth://totp/alumno@servidor%3Fsecret%3DNBGXKOCYHIYT4MHVDGKFNSVXLY%26issuer%3Dservidor>



Your new secret key is: NBGXKOCYHIYT4MHVDGKFNSVXLY

Enter code from app (-1 to skip):

```
pc-jsantos@DESKTOP-C1743AU:/mnt/c/Users/pc$ ssh alumno@192.168.56.101
Enter passphrase for key '/home/pc-jsantos/.ssh/id_ed25519':
Enter passphrase for key '/home/pc-jsantos/.ssh/id_ed25519':
Enter passphrase for key '/home/pc-jsantos/.ssh/id_ed25519':
(alumno@192.168.56.101) Password:
(alumno@192.168.56.101) Verification code:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-47-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

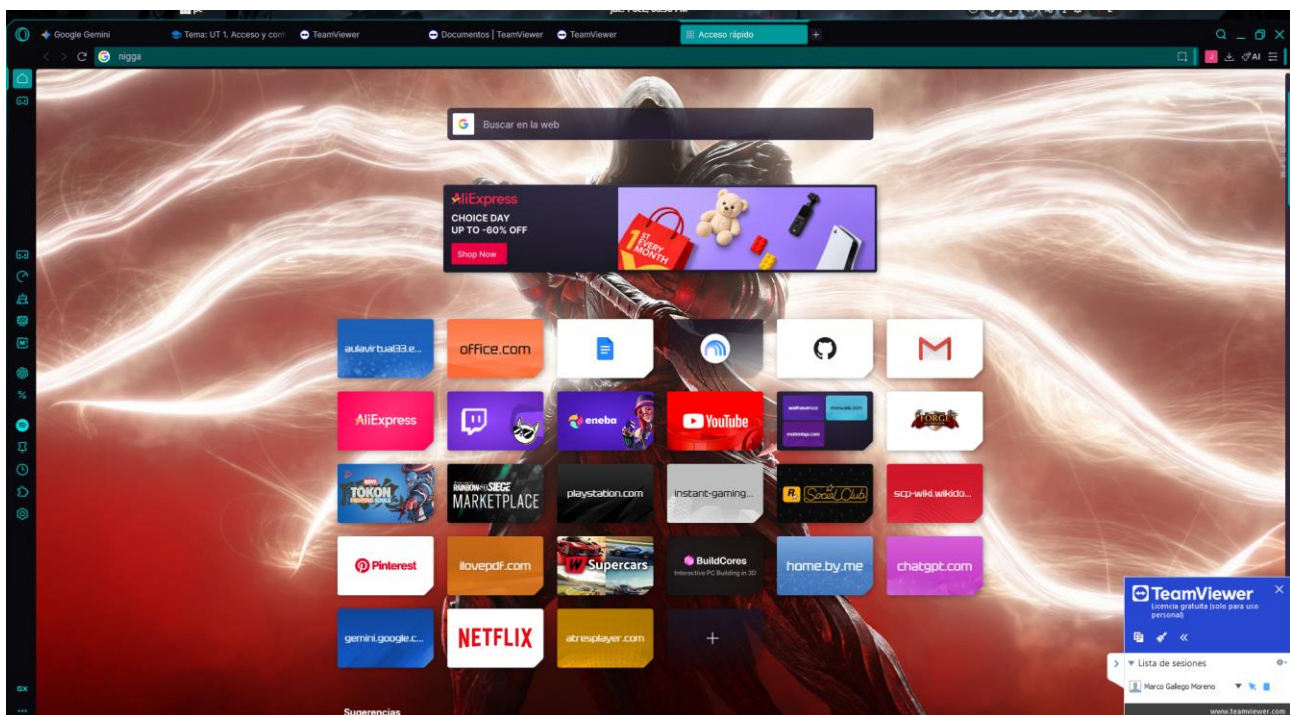
This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Oct  2 18:37:49 2026 from 192.168.56.1
alumno@servidor:~$
```

5. En parejas de clase instalad el teamviewer en casa y controlad el equipo de vuestro compañero.

Captura de pantalla de vuestro ordenador controlado por vuestro compañero



Captura de pantalla en que se muestra que controláis a vuestro compañero.

